CONVEXITY AND BOUNDARY POSITIVITY OF THE HESSIAN

NAM LE

Abstract. Let $C \subset \mathbb{R}^N$ be a closed convex set with nonempty interior, and let $f\colon C \to \mathbb{R}$ be convex and twice differentiable on C. It is standard that $\nabla^2 f(x) \succeq 0$ for every $x \in \operatorname{int}(C)$, since the usual open-set characterization of convexity by the positivity of the Hessian applies on the interior. The main issue is whether the same conclusion must remain true at boundary points when no continuity of the Hessian is assumed up to ∂C. We show that, under the relative notion of twice differentiability used here, convexity still forces $\nabla^2 f(x) \succeq 0$ at every point of C, including boundary points. We also show that this conclusion does not come from automatic C^2 regularity: there exist convex twice differentiable functions whose Hessian is not continuous.

1. Introduction

For convex functions defined on open subsets of $\mathbb{R}^N$, second-order information is well understood: if f is twice differentiable and convex on an open set U, then the Hessian satisfies

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in U. \tag{1}$$

See, for instance, Rockafellar [1970] or Hiriart-Urruty and Lemaréchal [2001] for standard background on convex functions and their second-order characterizations.

In the present note, the domain is not assumed open. Instead, we consider a closed convex set $C \subset \mathbb{R}^N$ with nonempty interior and a function $f\colon C \to \mathbb{R}$ that is convex on C.

The interior of the problem is immediate: restricting attention to $\operatorname{int}(C)$ gives the standard conclusion

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in \operatorname{int}(C). \tag{2}$$

The difficulty lies at boundary points. If one strengthens the regularity assumption and requires that f be of class C^2 on C, then the boundary question is again straightforward, because the Hessian can be obtained as a limit of interior Hessians. Without continuity of $\nabla^2 f$ up to the boundary, however, the situation is less clear.

The first main result of this note, stated as Theorem 4.3, shows that the boundary obstruction is illusory under the relative notion of twice differentiability adopted here: one still has

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in C. \tag{3}$$

The second main point is complementary. Proposition 5.1 shows that convexity plus twice differentiability does not imply continuity of the Hessian, even in one dimension. Thus the boundary positivity theorem is not merely a corollary of an automatic C^2 regularity principle.

There is also a geometric reason to separate these issues carefully. At a point $x \in \partial C$, convexity only tests the behavior of f along directions that keep one inside

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the set. Consequently, even if second-order information is available at x, one must distinguish between positivity of the quadratic form $h \mapsto \langle \nabla^2 f(x)h, h\rangle$ on all of $\mathbb{R}^N$ and positivity only on the cone of feasible directions determined by C at x. This distinction is one of the central themes in what follows. The next section records the precise relative second-order framework needed at boundary points, after which Section 3 reformulates the motivating questions in a way that leads directly to the proofs.

2. Background and Preliminaries

We fix a closed convex set $C \subset \mathbb{R}^N$ with $\operatorname{int}(C) \neq \varnothing$ and a function $f\colon C \to \mathbb{R}$.

2.1. **Convexity on a closed convex set.** The function f is said to be convex on C if

$$f((1-\alpha)x + \alpha y) \leq (1-\alpha)f(x) + \alpha f(y) \tag{4}$$

for every $x, y \in C$ and every $\alpha \in [0,1]$. Since C is convex, each segment $[x,y]$ is contained in C, so the restriction of f to any such segment is a one-variable convex function.

Two standard facts will be used repeatedly.

Proposition 2.1. *If f is convex on C, then f is continuous on $\operatorname{int}(C)$. Moreover, if f is differentiable on $\operatorname{int}(C)$, then ∇f is continuous on $\operatorname{int}(C)$.*

Proof. Fix $x_0 \in \operatorname{int}(C)$. Choose $r > 0$ such that

$$Q := x_0 + [-r, r]^N \subset \operatorname{int}(C), \tag{5}$$

and let V be the set of vertices of Q. Set

$$M := \max_{v \in V} f(v). \tag{6}$$

Since $Q = \operatorname{co}(V)$ and f is convex, we have

$$f(y) \leq M \qquad \forall y \in Q. \tag{7}$$

Now let

$$Q_{1/2} := x_0 + [-r/2, r/2]^N. \tag{8}$$

If $y \in Q_{1/2}$, define $z := 2x_0 - y$. Then $z \in Q$ and

$$x_0 = \frac{y+z}{2}. \tag{9}$$

By convexity,

$$f(x_0) \leq \frac{f(y)+f(z)}{2} \leq \frac{f(y)+M}{2}, \tag{10}$$

so

$$f(y) \geq 2f(x_0) - M \qquad \forall y \in Q_{1/2}. \tag{11}$$

Thus f is bounded on $Q_{1/2}$: there exists $B > 0$ such that

$$|f(y)| \leq B \qquad \forall y \in Q_{1/2}. \tag{12}$$

We next prove that f is Lipschitz on the smaller cube

$$Q_{1/4} := x_0 + [-r/4, r/4]^N. \tag{13}$$

Let $y, z \in Q_{1/4}$ with $y \neq z$, and define

$$t := \frac{r}{4\|z-y\|_\infty}, \qquad w := z + t(z-y). \tag{14}$$

Then

$$\|w - x_0\|_\infty \le \|z - x_0\|_\infty + t\|z - y\|_\infty \le \frac{r}{4} + \frac{r}{4} = \frac{r}{2}, \tag{15}$$

so $w \in Q_{1/2}$. Moreover,

$$w = (1+t)z - ty, \tag{16}$$

hence

$$z = \frac{t}{1+t}y + \frac{1}{1+t}w. \tag{17}$$

By convexity,

$$f(z) \le \frac{t}{1+t}f(y) + \frac{1}{1+t}f(w), \tag{18}$$

so

$$f(z) - f(y) \le \frac{f(w) - f(y)}{1+t} \le \frac{2B}{1+t}. \tag{19}$$

Since

$$\frac{1}{1+t} = \frac{1}{1 + \frac{r}{4\|z-y\|_\infty}} \le \frac{4\|z-y\|_\infty}{r}, \tag{20}$$

we obtain

$$f(z) - f(y) \le \frac{8B}{r}\|z - y\|_\infty. \tag{21}$$

Interchanging y and z gives

$$|f(z) - f(y)| \le \frac{8B}{r}\|z - y\|_\infty \qquad \forall y, z \in Q_{1/4}. \tag{22}$$

Thus f is continuous at x_0. Since x_0 was arbitrary, f is continuous on $\mathrm{int}(C)$.

Assume now that f is differentiable on $\mathrm{int}(C)$, and fix $x \in \mathrm{int}(C)$. By the first part, f is locally Lipschitz near x, so there exist $\rho > 0$ and $L > 0$ such that

$$B(x, \rho) \subset \mathrm{int}(C) \tag{23}$$

and

$$|f(y) - f(z)| \le L\|y - z\| \qquad \forall y, z \in B(x, \rho). \tag{24}$$

Let $y \in B(x, \rho/2)$ and let $u \in \mathbb{R}^N$ with $\|u\| = 1$. For all sufficiently small t, one has $y + tu \in B(x, \rho)$, hence

$$\left|\frac{f(y + tu) - f(y)}{t}\right| \le L. \tag{25}$$

Letting $t \to 0$ and using differentiability at y gives

$$|\langle \nabla f(y), u\rangle| \le L. \tag{26}$$

Taking the supremum over all unit vectors u, we obtain

$$\|\nabla f(y)\| \le L \qquad \forall y \in B(x, \rho/2). \tag{27}$$

Thus ∇f is locally bounded on $\mathrm{int}(C)$.

To prove continuity of ∇f at x, let $y_n \to x$ with $y_n \in \mathrm{int}(C)$. Discarding finitely many terms, we may assume $y_n \in B(x, \rho/2)$ for all n. Then $\{\nabla f(y_n)\}$ is bounded, so every subsequence has a convergent subsubsequence. Let

$$\nabla f(y_{n_k}) \to p. \tag{28}$$

Fix $z \in C$. For any $s \in (0, 1]$, convexity gives

$$f((1-s)y_{n_k} + sz) \le (1-s)f(y_{n_k}) + sf(z). \tag{29}$$

Rearranging,

$$\frac{f(y_{n_k} + s(z - y_{n_k})) - f(y_{n_k})}{s} \leq f(z) - f(y_{n_k}). \tag{30}$$

Letting $s \downarrow 0$ and using differentiability at y_{n_k}, we obtain

$$\langle \nabla f(y_{n_k}), z - y_{n_k} \rangle \leq f(z) - f(y_{n_k}), \tag{31}$$

that is,

$$f(z) \geq f(y_{n_k}) + \langle \nabla f(y_{n_k}), z - y_{n_k} \rangle. \tag{32}$$

Passing to the limit as $k \to \infty$ and using continuity of f at x gives

$$f(z) \geq f(x) + \langle p, z - x \rangle \qquad \forall z \in C. \tag{33}$$

Hence $p \in \partial f(x)$. Since f is differentiable at x, the subdifferential reduces to the singleton $\{\nabla f(x)\}$, so

$$p = \nabla f(x). \tag{34}$$

Thus every convergent subsubsequence of $\{\nabla f(y_n)\}$ has limit $\nabla f(x)$, and therefore

$$\nabla f(y_n) \to \nabla f(x). \tag{35}$$

Hence ∇f is continuous at x. Since x was arbitrary, ∇f is continuous on $\mathrm{int}(C)$. □

2.2. **Twice differentiability on C.** Because the domain is not open, the notion of twice differentiability at a boundary point must be stated explicitly. In this note, we say that f is twice differentiable at $x \in C$ if there exist a vector $\nabla f(x) \in \mathbb{R}^N$ and a symmetric matrix $\nabla^2 f(x) \in \mathbb{R}^{N \times N}$ such that

$$f(x + h) = f(x) + \langle \nabla f(x), h \rangle + \frac{1}{2} \langle \nabla^2 f(x) h, h \rangle + o(\|h\|^2) \tag{36}$$

as $h \to 0$ under the constraint that $x + h \in C$. Equivalently, the remainder is required to satisfy

$$f(x + h) - f(x) - \langle \nabla f(x), h \rangle - \frac{1}{2} \langle \nabla^2 f(x) h, h \rangle = o(\|h\|^2) \tag{37}$$

for admissible increments h with $x + h \in C$.

This is a relative notion of second-order differentiability along the set C. When $x \in \mathrm{int}(C)$, it coincides with the usual definition on an open neighborhood of x. At a boundary point, it captures second-order expansions only along directions that remain in the domain.

2.3. **Feasible directions and tangent cones.** For $x \in C$, the feasible-direction cone of C at x is

$$T_C(x) := \overline{\{t(y - x) : t \geq 0,\ y \in C\}}. \tag{38}$$

Equivalently, $v \in T_C(x)$ if there exist sequences $t_k \downarrow 0$ and $v_k \to v$ such that $x + t_k v_k \in C$ for all k. When $x \in \mathrm{int}(C)$, one has $T_C(x) = \mathbb{R}^N$; at a boundary point, this cone encodes the directions in which one can move while remaining in C; see Rockafellar and Wets [1998] for standard tangent-cone constructions in variational analysis.

This cone is relevant because convexity of f on C can only be tested along segments contained in C. Thus, at $x \in \partial C$, any immediate second-order consequence of convexity is expected to involve the quadratic form $h \mapsto \langle \nabla^2 f(x) h, h \rangle$ at least on $T_C(x)$, whereas positivity in arbitrary ambient directions is a stronger requirement.

2.4. **Standard second-order criterion on the interior.** The following is the classical open-set result applied to the interior of C.

Proposition 2.2. *Suppose that f is convex on C and twice differentiable on* $\operatorname{int}(C)$. *Then*

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in \operatorname{int}(C). \tag{39}$$

Proof. Fix $x \in \operatorname{int}(C)$ and $h \in \mathbb{R}^N$. Since x is an interior point of C, there exists $\varepsilon > 0$ such that

$$x + th \in C \qquad \forall t \in (-\varepsilon, \varepsilon). \tag{40}$$

Define

$$\phi(t) := f(x + th), \qquad t \in (-\varepsilon, \varepsilon). \tag{41}$$

Because f is convex on C, the function ϕ is convex on $(-\varepsilon, \varepsilon)$.

Since f is twice differentiable at x,

$$\phi(t) = \phi(0) + t\langle \nabla f(x), h\rangle + \frac{1}{2}t^2\langle \nabla^2 f(x)h, h\rangle + o(t^2) \qquad (t \to 0). \tag{42}$$

Applying this expansion with t and with $-t$, we get

$$\phi(-t) = \phi(0) - t\langle \nabla f(x), h\rangle + \frac{1}{2}t^2\langle \nabla^2 f(x)h, h\rangle + o(t^2). \tag{43}$$

Therefore,

$$\phi(t) + \phi(-t) - 2\phi(0) = t^2\langle \nabla^2 f(x)h, h\rangle + o(t^2). \tag{44}$$

Since ϕ is convex,

$$\phi(0) \le \frac{\phi(t) + \phi(-t)}{2}, \tag{45}$$

hence

$$0 \le \phi(t) + \phi(-t) - 2\phi(0). \tag{46}$$

Combining this with the previous expansion, we obtain

$$0 \le t^2\langle \nabla^2 f(x)h, h\rangle + o(t^2). \tag{47}$$

Dividing by $t^2 > 0$ and letting $t \to 0$ yields

$$\langle \nabla^2 f(x)h, h\rangle \ge 0. \tag{48}$$

Since $h \in \mathbb{R}^N$ was arbitrary, it follows that

$$\nabla^2 f(x) \succeq 0. \tag{49}$$

Because $x \in \operatorname{int}(C)$ was arbitrary, the proof is complete. □

Remark 2.3. If one additionally assumes that f is of class C^2 on the whole of C in the sense that the Hessian extends continuously to boundary points, then the interior positivity immediately passes to every $x \in C$ by taking limits from $\operatorname{int}(C)$.

3. Motivation and Main Questions

We can now formulate the motivating questions in the language fixed above.

Question 1 (Boundary positivity of the Hessian)**.** Let $C \subset \mathbb{R}^N$ be a closed convex set with nonempty interior, and let $f\colon C \to \mathbb{R}$ be convex and twice differentiable on C in the relative sense above. Must one have

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in C? \tag{50}$$

Equivalently, does convexity on C force the Hessian at a boundary point to be positive semidefinite on all ambient directions of $\mathbb{R}^N$?

This question is nontrivial only on ∂C, since the conclusion is already known on $\operatorname{int}(C)$ by Proposition 2.2. It is also already settled under the stronger assumption that f is C^2 on C by Remark 2.3. The point of Theorem 4.3 is that no such continuity assumption is actually needed.

Question 2 (Automatic continuity of the Hessian)**.** If f is convex and twice differentiable on C, must the mapping $x \mapsto \nabla^2 f(x)$ be continuous, at least on compact subsets of $\operatorname{int}(C)$ or up to the boundary under the relative notion of twice differentiability?

An affirmative answer to the second question would give a direct positive answer to the first one, but it is not clear a priori that convexity plus existence of pointwise second derivatives should force such continuity.

There is also a weaker boundary formulation which is geometrically natural.

Question 3 (Positivity on feasible directions)**.** For $x \in \partial C$, does convexity at least imply that

$$\langle \nabla^2 f(x)h, h\rangle \geq 0 \qquad \forall h \in T_C(x)? \tag{51}$$

If so, to what extent can this feasible-direction positivity be upgraded to full positive semidefiniteness on $\mathbb{R}^N$?

This reformulation isolates the geometric core of the problem. Convexity of the restriction of f to segments in C naturally gives one-dimensional second-order inequalities only in directions allowed by the set. The passage from such restricted inequalities to an ambient matrix inequality is exactly the point that must be clarified. The results of the next section answer Question 3 first and then upgrade the conclusion to the full ambient statement in Question 1.

Remark 3.1. The logical structure of the paper is now as follows. Section 4 first proves positivity on feasible directions by one-dimensional convexity along admissible segments and then derives the full matrix inequality. Section 5 then shows, by explicit counterexample, that the Hessian continuity question from Question 2 has a negative answer.

4. Boundary Second-Order Positivity

We begin with the cone of directions that are realized by genuine admissible segments. This auxiliary cone allows one to extract one-dimensional convexity information before passing to the ambient quadratic form.

For $x \in C$, let

$$A_C(x) := \{h \in \mathbb{R}^N : x + th \in C \text{ for some } t > 0\}. \tag{52}$$

Because C is convex and $x \in C$, this set is a convex cone and satisfies

$$T_C(x) = \overline{A_C(x)}. \tag{53}$$

Moreover, $A_C(x)$ has nonempty interior. Indeed, if $z \in \operatorname{int}(C)$ and $B(z,r) \subset C$, then

$$(z-x) + B(0,r) \subset A_C(x), \tag{54}$$

because $x + 1 \cdot ((z-x)+u) = z + u \in C$ for every $u \in B(0,r)$.

We first extract the boundary information that comes directly from convexity along admissible rays.

Proposition 4.1. *Let $x \in C$. If f is convex on C and twice differentiable at x in the relative sense, then*

$$\langle \nabla^2 f(x)h, h\rangle \geq 0 \qquad \forall h \in T_C(x). \tag{55}$$

Proof. We first consider $h \in A_C(x)$. By definition, there exists $t_0 > 0$ such that $x + t_0 h \in C$. Since C is convex and $x \in C$, it follows that $x + th \in C$ for every $t \in [0, t_0]$. Define

$$\phi_h(t) := f(x+th), \qquad 0 \leq t \leq t_0. \tag{56}$$

Then ϕ_h is convex on $[0, t_0]$, because it is the restriction of f to an admissible line segment.

The second-order expansion of f at x gives

$$\phi_h(t) = \phi_h(0) + t\,\langle \nabla f(x), h\rangle + \frac{1}{2}t^2\,\langle \nabla^2 f(x)h, h\rangle + o(t^2) \qquad (t \downarrow 0). \tag{57}$$

Since ϕ_h is convex, the midpoint inequality yields

$$0 \leq \phi_h(t) + \phi_h(0) - 2\phi_h(t/2) \tag{58}$$

for every sufficiently small $t > 0$. Substituting the expansion above gives

$$0 \leq \frac{1}{4}t^2\,\langle \nabla^2 f(x)h, h\rangle + o(t^2). \tag{59}$$

Dividing by t^2 and letting $t \downarrow 0$, we obtain

$$\langle \nabla^2 f(x)h, h\rangle \geq 0 \qquad \forall h \in A_C(x). \tag{60}$$

Now let $h \in T_C(x) = \overline{A_C(x)}$. Choose $h_k \in A_C(x)$ with $h_k \to h$. Since $v \mapsto \langle \nabla^2 f(x)v, v\rangle$ is continuous, we may pass to the limit and conclude that

$$\langle \nabla^2 f(x)h, h\rangle \geq 0. \tag{61}$$

This proves the claim. □

The previous proposition shows that convexity always forces nonnegativity on the tangent cone. The next result shows that, under the present notion of twice differentiability, one in fact obtains positivity on all ambient directions.

Lemma 4.2. *Let $A \subset \mathbb{R}^N$ be a convex set with nonempty interior, and let $q(h) = \frac{1}{2}\langle Hh, h\rangle$ be a quadratic form associated with a symmetric matrix H. If q is convex on A, then $H \succeq 0$.*

Proof. Assume for contradiction that H is not positive semidefinite. Then there exists $v \in \mathbb{R}^N$ such that $\langle Hv, v\rangle < 0$. Choose $u \in \operatorname{int}(A)$. For sufficiently small $\varepsilon > 0$, both $u+\varepsilon v$ and $u-\varepsilon v$ belong to A. By convexity of q on the segment joining these two points,

$$0 \le q(u+\varepsilon v) + q(u-\varepsilon v) - 2q(u) = \varepsilon^2\langle Hv, v\rangle < 0, \tag{62}$$

which is impossible. □

Theorem 4.3. *Let $C \subset \mathbb{R}^N$ be a closed convex set with nonempty interior, and let $f\colon C \to \mathbb{R}$ be convex and twice differentiable on C in the relative sense of the previous section. Then*

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in C. \tag{63}$$

Proof. Fix $x \in C$ and write

$$q_x(h) := \frac{1}{2}\langle \nabla^2 f(x)h, h\rangle. \tag{64}$$

For $s > 0$, define the scaled admissible set

$$D_s(x) := \{h \in \mathbb{R}^N : x + sh \in C\} \tag{65}$$

and the scaled difference quotient

$$g_{x,s}(h) := \frac{f(x+sh) - f(x) - s\langle \nabla f(x), h\rangle}{s^2}, \qquad h \in D_s(x). \tag{66}$$

Each $D_s(x)$ is convex, and each $g_{x,s}$ is convex on $D_s(x)$ because f is convex on C. We claim that q_x is convex on $A_C(x)$. Let $u, v \in A_C(x)$ and let $\lambda \in [0,1]$. By definition of $A_C(x)$, there exist $t_u, t_v > 0$ such that $x + t_u u \in C$ and $x + t_v v \in C$. Set $s_0 := \min\{t_u, t_v\}$. For every $s \in (0, s_0]$, convexity of C implies $u, v \in D_s(x)$, hence also $(1-\lambda)u + \lambda v \in D_s(x)$. Therefore

$$g_{x,s}((1-\lambda)u + \lambda v) \le (1-\lambda)g_{x,s}(u) + \lambda g_{x,s}(v) \tag{67}$$

for all $s \in (0, s_0]$.

Since $u, v, (1-\lambda)u+\lambda v \in A_C(x)$, the relative second-order expansion at x implies

$$g_{x,s}(u) \to q_x(u), \qquad g_{x,s}(v) \to q_x(v), \qquad g_{x,s}((1-\lambda)u + \lambda v) \to q_x((1-\lambda)u + \lambda v) \tag{68}$$

as $s \downarrow 0$. Passing to the limit in the convexity inequality above, we obtain

$$q_x((1-\lambda)u + \lambda v) \le (1-\lambda)q_x(u) + \lambda q_x(v). \tag{69}$$

Thus q_x is convex on the convex set $A_C(x)$.

As noted above, $A_C(x)$ has nonempty interior. Lemma 4.2 therefore implies that $\nabla^2 f(x) \succeq 0$. Since $x \in C$ was arbitrary, the proof is complete. □

Remark 4.4. Theorem 4.3 is stronger than the tangent-cone statement in Proposition 4.1. Proposition 4.1 uses only convexity of the one-variable restrictions $t \mapsto f(x+th)$ along admissible rays, whereas Theorem 4.3 also exploits convexity among different admissible directions through the scaled functions $g_{x,s}$. This additional interaction is what upgrades feasible-direction nonnegativity to a full ambient matrix inequality.

5. Failure of Automatic Hessian Continuity

Theorem 4.3 settles the boundary positivity question without any appeal to continuity of the Hessian. It remains to separate that positivity phenomenon from the regularity question raised earlier. The next proposition shows that convexity plus twice differentiability does not force C^2 regularity.

Proposition 5.1. *There exists a convex function $F\colon \mathbb{R} \to \mathbb{R}$ which is twice differentiable at every point, but whose second derivative is not continuous. Consequently, for every $N \geq 1$ there exists a convex twice differentiable function on $\mathbb{R}^N$ whose Hessian is not continuous.*

Proof. Define

$$g(x) := \begin{cases} 2 + \sin(1/x), & x \neq 0, \\ 2, & x = 0, \end{cases} \tag{70}$$

and set

$$F(x) := \int_0^x \left(\int_0^t g(s)\, ds \right) dt. \tag{71}$$

Then

$$F'(x) = \int_0^x g(s)\, ds. \tag{72}$$

Since $g \geq 1$, we have

$$F'(y) - F'(x) = \int_x^y g(s)\, ds \geq y - x \qquad \forall x < y, \tag{73}$$

so F' is increasing and therefore F is convex.

For every $x \neq 0$, the function g is continuous at x, hence the classical fundamental theorem of calculus gives

$$F''(x) = g(x) = 2 + \sin(1/x). \tag{74}$$

It remains to check twice differentiability at the origin. Since $F'(0) = 0$, we compute

$$\frac{F'(h) - F'(0)}{h} = \frac{1}{h} \int_0^h g(s)\, ds = 2 + \frac{1}{h} \int_0^h \sin(1/s)\, ds. \tag{75}$$

For $h > 0$, the substitution $u = 1/s$ gives

$$\int_0^h \sin(1/s)\, ds = \int_{1/h}^\infty \frac{\sin u}{u^2}\, du. \tag{76}$$

An integration by parts shows that the right-hand side is $O(h^2)$ as $h \downarrow 0$, and the same estimate holds for $h \uparrow 0$. Therefore

$$\frac{1}{h} \int_0^h \sin(1/s)\, ds \to 0, \tag{77}$$

so $F''(0) = 2$ exists. Thus F is twice differentiable at every point, but $F''(x) = 2 + \sin(1/x)$ for $x \neq 0$, which is not continuous at 0.

For $N \geq 2$, one may define

$$f(x_1, \ldots, x_N) := F(x_1) + x_2^2 + \cdots + x_N^2. \tag{78}$$

Then f is convex and twice differentiable on $\mathbb{R}^N$, while its Hessian is not continuous at the origin. $\square$

Remark 5.2. Proposition 5.1 shows that Theorem 4.3 is genuinely stronger than the easy C^2 argument based on continuity of the Hessian. The positivity conclusion survives at boundary points even though automatic Hessian continuity fails.

6. Conclusion

Two conclusions emerge from the preceding analysis. First, under the relative notion of twice differentiability adopted in this note, convexity on a closed convex set with nonempty interior still forces the ambient Hessian to satisfy

$$\nabla^2 f(x) \succeq 0 \qquad \forall x \in C,$$

including at boundary points. Second, this positive conclusion is not a byproduct of automatic C^2 regularity: convex twice differentiable functions may still have discontinuous Hessian.

In this sense, the paper separates two issues that are easy to conflate. Boundary positivity is a geometric consequence of convexity together with a relative second-order expansion on the domain, whereas continuity of the Hessian is an additional regularity property that can fail. A natural direction for further study is to compare this conclusion with what happens under alternative notions of second-order differentiability on non-open sets.

References

Jean-Baptiste Hiriart-Urruty and Claude Lemaréchal. *Fundamentals of Convex Analysis*. Springer, Berlin, 2001.

R. Tyrrell Rockafellar. *Convex Analysis*, volume 28 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1970.

R. Tyrrell Rockafellar and Roger J.-B. Wets. *Variational Analysis*, volume 317 of *Grundlehren der Mathematischen Wissenschaften*. Springer, Berlin, 1998.

Department of Computer Science, Faculty of Information Technology, University of Science, Vietnam National University, 227 Nguyen Van Cu Street, Cho Quan Ward, Ho Chi Minh City

Email address: lnnam@hcmus.edu.vn